Automatic Tool Change System for Uniquely Functional End Effectors for Robot Arms
Little Daisies MKII

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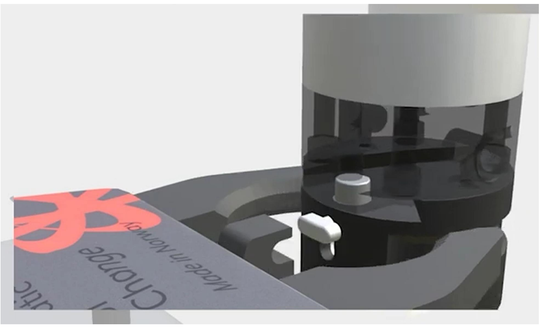
Automatic Tool Change System

Objective: Design and manufacture a reliable tool changer for robot arms with the following properties,

- Fully Automatic
- Manual Operation if Desired *
- Connects Arm and End Effector Electronically
- Robust to Positional Errors
- Resistant to Wear & Plastic Deformation



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A Library of End Effectors

Objective: Design and manufacture 2-3 end effectors with unique functionalities and purposes. Some potential end effectors could be:

Cardboard Model Maker

- Suction Holder
- Glue/Pritt Holder
- Cutter

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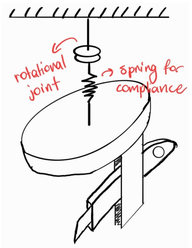

Table Tidyer

- Compliant Gripper
- Table Cleaner
- Cloth Holder *

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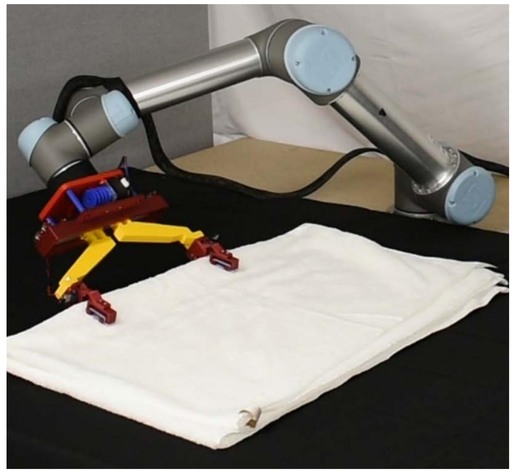
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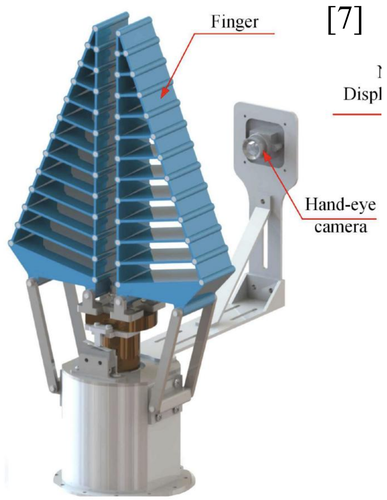

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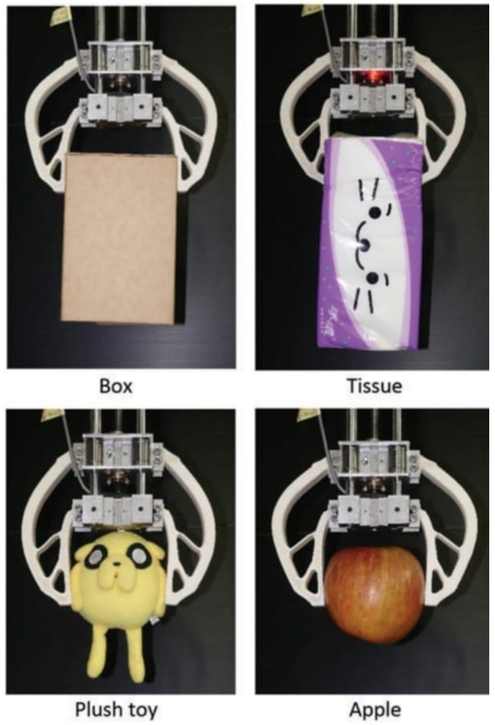
Testing/Demonstration

- Robot arm path simulation
- End effector simulation *
- Hardcoded serial tasks / manual operation
- Automated simpler tasks * (potentially with overhead or tool mounted cameras)

References

[1] https://www.youtube.com/watch?v=DKoga75gqLg&list=PLPCmeat3ZauMB9G-j1zMtYtTFtzrOmsXM&index=10
[2] https://www.youtube.com/watch?v=B2dpcNf5LFg&list=PLPCmeat3ZauMB9G-j1zMtYtTFtzrOmsXM&index=10
[3] https://www.youtube.com/watch?v=IWDu6bApWZE&list=PLPCmeat3ZauMB9G-j1zMtYtTFtzrOmsXM&index=12
[4] https://www.youtube.com/watch?v=3M4gb9KVHWo&list=PLPCmeat3ZauMB9G-j1zMtYtTFtzrOmsXM&index=7
[5] Craftsman Auto Buffer Polisher
[6] Lee, Dongmyoung & Chen, Wei & Chen, Xiaoshuai & Rojas, Nicolás. (2024). G.O.G: A Versatile Gripper-On-Gripper Design for Bimanual Cloth Manipulation with a Single Robotic Arm. IEEE Robotics and Automation Letters. PP. 1-8. 10.1109/LRA.2024.3357028.
{7] W. Xu, H. Zhang, H. Yuan and B. Liang, "A Compliant Adaptive Gripper and Its Intrinsic Force Sensing Method," in IEEE Transactions on Robotics, vol. 37, no. 5, pp. 1584-1603, Oct. 2021, doi: 10.1109/TRO.2021.3060971.
[8] Lee, D., Chen, W., Chen, X., & Rojas, N. (2024). Gog: A versatile gripper-on-gripper design for bimanual cloth manipulation with a single robotic arm. *IEEE Robotics and Automation Letters*, 9(3), 2415-2422.
[9] https://tr.pinterest.com/pin/140806233672011/



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* Project design specifications/requirements are subject to change and are here as means of suggestions/ideas

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CNC RETROFITTING

ME462

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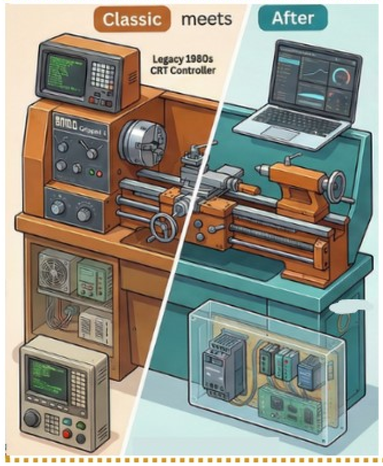
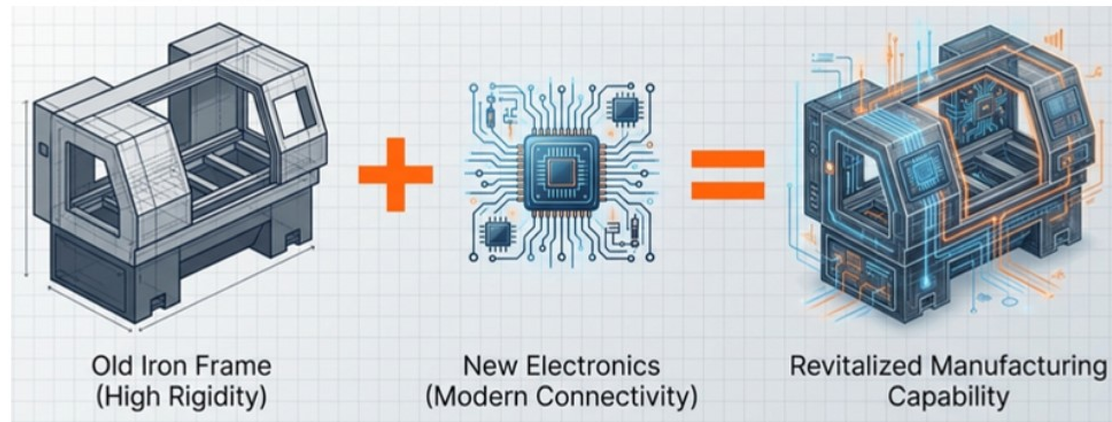


What is CNC Retrofitting?

The process of modernizing an existing machine by upgrading its control, electrical, software, and sensing systems to restore operation and improve performance.



Why Retrofit? (The Strategic Value)

Cost-Effectiveness & Sustainability

Retrofitting reuses mechanically solid machine structures, such as high-quality cast iron frames, reducing waste and saving significantly compared to buying new hardware.

Extended Machine Life

By updating obsolete power electronics and motor drivers, the machine's operational lifespan is significantly extended with improved maintainability.

Educational and Research Value

Retrofitting provides a transparent look at control theory and mechatronics, allowing for customized software stacks and open-souree integration.

CNC Retrofitting Process Workflow

01 MACHINE ASSESSMENT
Evaluating mechanical health and identifying surviving electrical components.

02 SCOPE & DOCUMENTATION
Defining retrofit goals and performing reverse engineering on legacy wiring and motor interfaces.

03 ARCHITECTURE PLANNING
Designing the generic controller hardware layout and selecting modern motor drivers.

04 ELECTRICAL UPGRADES
Wiring organization and installation of new power electronics and safety circuits.

05 SYSTEM INTEGRATION
Linking motion control, sensors, and UO modules into the unified platform.

06 SOFTWARE & LOGIC
Installation of the control software stack and configuration of machine-specific parameters.

07 SAFETY & COMMISSIONING
Implementing hardware interlocks and performing initial motion testing and sensor calibration.

08 FINAL OPERATION
Validation of the lathe and mill through precision machining teats and O-code execution.

Machine Adaptation Layer

Lathe Module
Specialized translation of generic commands to lathe-specific movements.

Mill Module
Specialized translation of generic commands to mill-specific movements.

Common Challanges

- Missing original documentation
- Electrical noise/signal interference
- Complex safety integration can extend timelines

Mini Mini Robotların Şirin Dünyası

Anatomy of workers

Capabilities:
- Lifting
- Linear motion and steering

Microstepper with SCREW:
- Cheap
- Small
- Can Lift

A Microcontroller
- Micro (optional)
- Preferably one that can process very basic video
- Must have Wifi to communicate with mother and other childs.
- Can utilize 1 camera 2 motor and couple other sensors

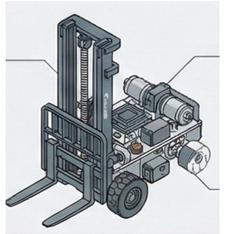
Miniature Ball Wheel
- Miniature
- Rotates freely

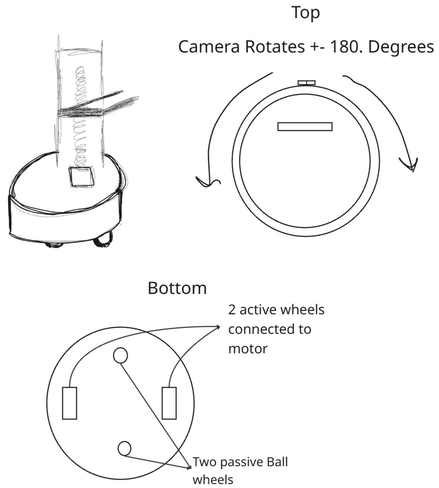
A micro Camera
- Compatible with microprocessor used
- They are very expensive
- May exterminate the need for overhead camera
- Without overhead camera will make things interasting
- Can follow line, recognize markers, recognize other forklifts

Tiny Dc Motor
- Tiny
- 2 of them needed for linear motion and steering

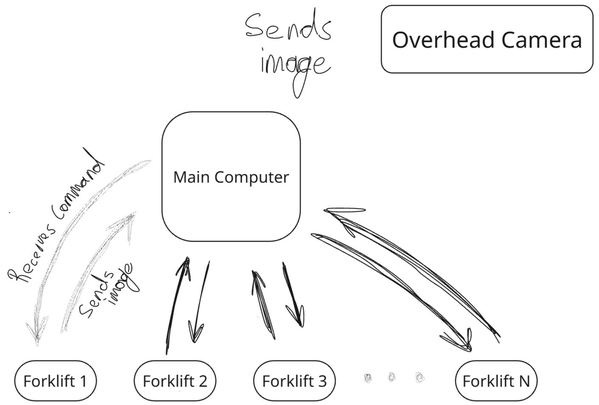
Smaller Version of this
- For sensing the obstacles that are in the back

Some Concepts

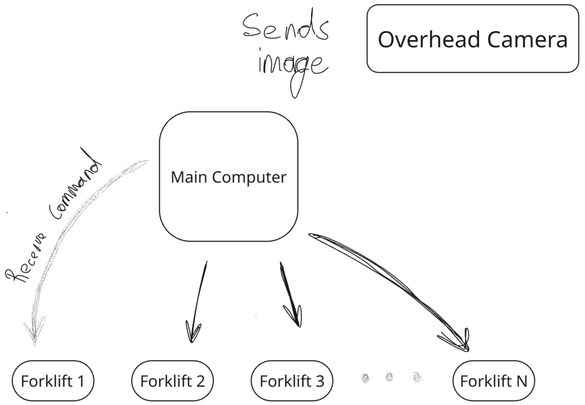




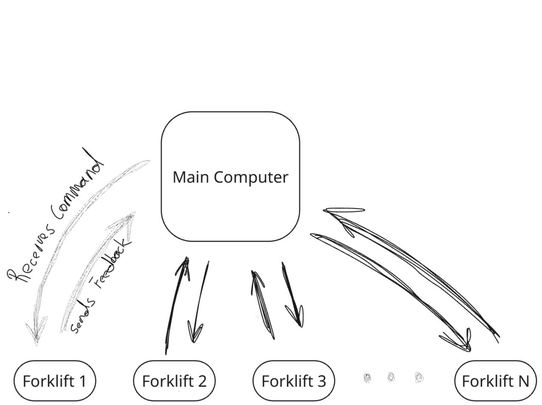
Communication Structre



System with the overhead Camera, single image processing:
- Overhead camera sends video feed to main computer for analysis
- All the worker forklifts send their own video feed to mother. The necessity of the individual cameras are in doubt in this case.
- Mother computer processes the provided feeds and send commands to workers. This commands include the goal and the movement commands.



System with the overhead Camera, local image processing:
- Overhead camera sends video feed to main computer for analysis
- Workers analyses their own video feed and decide their own movements.
- Mother computer processes the overhead video feed and follows the workers. Thus feedback from the workers are not necessary.



System without the overhead Camera, local image processing:
- All the worker forklifts process their own feed. They send the their information position speed status etc to mother.
- Mother blindly controls the overall order of the warehouse, send the commands to workers for goal destination.